ON-THE-JOB TRAINING (OJT) ASSESSMENT RUBRIC

60% of Final Grade • Joint assessment by Student (self) and Supervisor

SECTION 1 Student & Placement Information

Student Name	Tan Chai Beng
Programme / Course	NSEP - AI ROBOTICS
Company Name	LogikHaus
Position / Role	*AI & Robotic team engineer*
Department	Logikhaus Kinematics, Robotics & AI Modelling Group
Supervisor Name	Daniel Kho
Supervisor Position	CTO
Supervisor Email	daniel.kho@logik.haus
Supervisor Phone	6588740726
Placement Period	15 June 2026 – 14 October 2026

SECTION 2 Work Description (to be filled by Student)

Brief Description of Company / Organisation

What does the company do? Industry, size, main products/services.

LogikHaus Sdn. Bhd. is a microelectronics design and engineering company working in the semiconductor and electronics industry. The company provides FPGA and ASIC design services, digital hardware and IP cores, and engineering solutions for areas such as video and image processing, digital signal processing (DSP), AI, communications, and robotics. The company operates with a specialist engineering team and has operations in Malaysia and Singapore.

Summary of Your Role & Responsibilities

Describe your day-to-day tasks, duties, and areas of responsibility.

I work on an AI-on-FPGA bird detection project at LogikHaus. My role covers the whole pipeline: preparing training data, training the model, running it on the company's FPGA hardware, and checking that the hardware behaves as expected.

My day-to-day tasks and responsibilities include:

Dataset preparation: organising labelled bird images into training, validation and test sets, and testing image pre-processing such as Sobel edge detection.
Model training and evaluation: training YOLO object-detection models (YOLOv3-tiny and tiny-YOLOv2) with PyTorch, Ultralytics and Darknet on CPU and GPU, and comparing runs using precision, recall and mAP.
Testing: running models on new photos and camera pictures, and finding where they are weak.
Preparing models for hardware: converting trained models into the exact FP16 weight files the board's AI accelerator loads.
Board testing:
programming the Xilinx Artix-7 board with Vivado through the remote lab
uploading weights and images with the board dashboard
checking the board's output against my software model of the hardware
Reproducibility and documentation: keeping my work in shared folder and writing step-by-step guides so others can repeat each stage.

Key Projects / Assignments Undertaken

List and briefly describe the main projects or assignments you worked on.

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Bird detection model (YOLOv3-tiny): I built a bird detector from a labelled dataset of 280 training, 82 validation and 40 test images. I trained and compared several models, first on CPU and later on a GPU. I also tested whether Sobel edge pre-processing improves accuracy over normal colour (RGB) images. I then used the model to detect birds in new photos.

Model export and hardware assessment: I converted the trained model to ONNX and wrote a tool that exports its weights to a half-precision (FP16) binary file for an FPGA accelerator. I also compared ways to run the model on the target board: Xilinx Vitis-AI, FINN, porting an existing accelerator, or using a separate edge device.

tiny-YOLOv2 on Darknet (Linux server): I ran the Darknet tiny-YOLOv2 model on a Rocky Linux server and exported its weights to FP16, folding the batch-normalisation into the weights. I checked the export against my own NumPy version of the network, and prepared the dataset and settings for training it on birds.

FPGA AI core analysis:

I investigated the AI accelerator in the company's system-on-chip, running on a Xilinx Artix-7 (Nexys Video) board. The accelerator is a port of HWAC, an open-source tiny-YOLOv2 design.
I traced the board's weight file back to its published source and worked out how the accelerator stores its weights in memory.
I built a software model of the board that predicts which box it will draw.
I found, in the hardware design, why the board only ever shows one box.
Bird-trained model on the FPGA board: I trained a new tiny-YOLOv2 model to find birds and ignore people, logos and screen icons, built to the exact shape the hardware requires. I packed it into the board's weight and padding files and loaded it onto the board with the updated SoC build and web dashboard. I then tested it on real pictures through the camera, comparing each result with my software model's prediction.

Skills & Tools Used

Technical skills, software, frameworks, methodologies, or equipment you applied.

Programming: Python (NumPy, OpenCV, Pillow), Bash scripting, reading Verilog hardware source
AI / machine learning:
PyTorch, Ultralytics YOLO, Darknet, ONNX
transfer learning, data augmentation, pseudo-labelling, training with negative examples (people, logos, icons)
model evaluation (precision, recall, IoU, mAP), FP16 conversion, batch-normalisation folding
Computer vision: object detection (YOLOv2, YOLOv3); image pre-processing (Sobel edge detection, letterboxing, colour-channel order)
Hardware / FPGA:
Xilinx Artix-7 (Nexys Video), Vivado Hardware Manager, the HWAC accelerator
binary weight and image formats
loading files into board memory through the board dashboards
camera-based testing, remote lab access
Computing platforms: NVIDIA GPU training (RTX 5050, CUDA), Rocky Linux, Windows, macOS, Python virtual environments
Software tools: VS Code,
Methods:
reproducible experiments (fixed random seeds, recorded settings, checkpoint backups)
checking results against a reference (file comparison, a software model that predicts the hardware's output)
test planning and technical documentation

Key Achievements & Contributions

What did you deliver, improve, or contribute to during the placement?

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Accurate bird detector: I delivered a YOLOv3-tiny model with 90.5% precision, 84.9% recall and an mAP50 of 0.875 on the validation set (82 images, 201 birds).

Faster, reproducible training: I moved training from CPU to GPU. A full 100-epoch run now takes about 12 minutes, with results matching the earlier CPU runs. I also recovered the settings and scores of past runs from their saved model files.

Evidence-based pre-processing decision: I showed that Sobel edge pre-processing gave no real gain over plain RGB images (mAP50 0.859 vs 0.854). This supported keeping the simpler RGB approach.

Verified export tools: I built FP16 export tools for YOLOv3-tiny and tiny-YOLOv2. The tiny-YOLOv2 export gave the same detections as Darknet itself.

Found what the board was really running: I proved, by exact file comparison, that the board was using HWAC's published drone-trained weights rather than a bird model. This explained why it also drew boxes around people and logos.

Enabled custom models on the hardware: I worked out the accelerator's weight memory layout, which had blocked earlier attempts. With it, my software rebuild of the board's network found a test eagle with 0.905 confidence, and its box overlapped the true position with an IoU of 0.84.

Bird-specific weights: I used my YOLOv3 model to label 1,921 extra bird photos automatically. The new model improved on the stock model in every test:

Its top box landed on a bird in 93% of test images, compared with 65% for the stock model.
Confident boxes (score above 0.5) on photos of people dropped from 5% to 1%.
AP50 rose from 0.12 to 0.46.
First custom model running on the board: the board boxed the birds my software model predicted.

In a garden scene, my model's top box scored 0.69, while the stock model's best guess scored only 0.04.
In a scene with five geese, my model scored every goose 0.99–1.00, and the board boxed the one it ranked first.
This confirmed that the whole conversion process works on the real hardware.

Found the single-box limitation: I traced, in the accelerator's hardware design, why the board only draws one box. It keeps only the highest-scoring candidate and has no confidence threshold. I identified where the team could change the hardware to add a threshold or show more boxes.

Test tools and documentation: I wrote a tool that predicts the board's box for any photo or camera screenshot. I also made test images with expected results and step-by-step guides for each stage.

SECTION 3 Rubric Scoring Guide

Beginning	Developing	Proficient	Advanced
0–2 Little or no evidence; major gaps in understanding or application.	3–5 Partial evidence; inconsistent application or limited depth.	6–8 Consistent, reliable evidence; meets expected standard.	9–10 Exceptional, independent; exceeds expectations with insight.

SECTION 4 MLO1 — Application of Core Theoretical & Technical Knowledge in the Workplace

Focus: Connecting academic knowledge to the workplace.

#	Criterion	Student Self-Score (0-10)	Supervisor Score (0-10)	Average (auto)
1	Understands the technical concepts needed for the job	9.5		9.50
2	Turns instructions or requirements into a clear plan of action	9.5		9.50
3	Produces good quality work using the required tools and methods	9.5		9.50
4	Picks up new tools or skills when the work calls for it	9.5		9.50
5	Thinks carefully before making technical choices (weighs options based on factors like cost, quality, time, etc)	9.5		9.50
6	Follows proper work practices and uses tools the right way	9.5		9.50
7	Solves problems step by step instead of guessing	9.5		9.50
8	Connects what they learned in class to what they do at work	9.5		9.50

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MLO1 Subtotal — Average score across 8 criteria (out of 10)	**9.50**		**9.50**

SECTION 5 MLO4 — Reflection on Personal Growth & Adaptation to Workplace Cultures

Focus: Demonstrates personal development and ability to adapt to organisational norms, values, and working styles.

#	Criterion	Student Self-Score (0-10)	Supervisor Score (0-10)	Average (auto)
1	Reflects critically on workplace challenges and lessons learned	9.5		9.50
2	Demonstrates use of critical thinking to propose or consider alternative approaches	9.5		9.50
3	Shows evidence of self-directed learning and a proactive attitude toward new tasks	9.5		9.50
4	Acts on feedback to improve performance or understanding	9.5		9.50
5	Reflects on personal growth, strengths, and areas for improvement	9.5		9.50
6	Adapts to workplace culture and norms professionally	9.5		9.50
7	Communicates insights and learning in a clear, reflective, and professional manner	9.5		9.50
8	Demonstrates resilience, initiative, and overall professional maturity	9.5		9.50
MLO4 Subtotal — Average score across 8 criteria (out of 10)		**9.50**		**9.50**

SECTION 6 Final Score Summary (auto-calculated)

Component	Weight	Combined Avg (out of 10)	
MLO1 — Application of Knowledge		9.50	
MLO4 — Reflection & Adaptation		9.50	
Subtotal		19.00	
OJT TRAINING REPORT SCORE (the rubric portion of the 60% OJT component)			**57 / 60**

Note: The remaining 10% of the OJT component is awarded separately based on Attendance records.

SECTION 7 Qualitative Comments

Student's Self-Reflection

Key learnings, challenges faced, and how the placement contributed to personal and professional growth.

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Key learnings

This placement showed me how an AI model goes from training to running on real hardware. Training an accurate model is only the first step. The model then has to fit the hardware's fixed design, number format and memory layout, and every conversion can introduce errors. I learned to check each step against a reference instead of assuming it worked. I compared files, rebuilt the network in software, and predicted the board's result before testing it. That approach paid off when the board boxed the same bird my software model predicted.

Testing with real camera pictures taught me that training data must match real conditions. My model was strong on birds in the sky but weaker on birds perched among leaves and flowers, because most of its training images showed flying birds. I also learned the value of reproducibility: recording settings, backing up models and using version control, so results can be repeated later on a different machine.

Challenges faced

The biggest challenge was the accelerator's weight format, which had very little documentation. My first assumptions were wrong and gave results that made no sense, and finding the correct layout took careful, step-by-step investigation. Hardware testing was slow and done through a remote lab, so I learned to plan tests carefully and to check as much as possible in software first.

Some limits turned out to be in the hardware itself, such as the board showing only one box. This taught me to recognise when a problem needs the hardware team rather than more training. There were smaller challenges too:

A small dataset: I added more images by labelling them automatically.
Slow CPU training: I moved training to a GPU.
Software environment problems across macOS, Windows and Linux: these taught me to debug systematically.

Personal and professional growth

The placement made me more independent and persistent. Many problems had no ready-made answer, so I broke them into small, testable steps. I became more careful about evidence, and I now report results with numbers and proof rather than impressions. Working with colleagues responsible for the hardware and the board software showed me the value of clear communication and good documentation. I also learned to use AI coding tools productively while checking their output against real results. Overall, the placement strengthened my technical skills and my confidence with unfamiliar engineering problems. It has also increased my interest in edge AI and in designing hardware and software together.

Supervisor's Comments

Overall assessment of the student's performance, strengths, and recommended areas for development.

Tan Chai Beng worked on running a bird-detection AI model on the company's FPGA platform and performed well throughout the placement. Tan Chai Beng took the work from dataset preparation and model training all the way to a bird-trained model running on the board's AI accelerator, with the results confirmed on the hardware. Along the way, Tan Chai Beng solved a key technical problem, the accelerator's undocumented weight layout, which had been blocking progress.

Strengths

Strong problem-solving and persistence on difficult, poorly documented problems
A careful, evidence-based approach: results are verified and backed by measurements
Learns new tools and platforms quickly (PyTorch, Darknet, Linux, Vivado, GPU training)
Clear documentation that lets other team members repeat the work
Areas for development

Testing on the physical hardware earlier and more often, rather than relying mainly on software checks
Deeper knowledge of FPGA design (RTL and Vivado), so that hardware-side improvements such as showing more than one box can be tackled directly
Better organisation of project files, with fewer duplicate copies of folders and versions
I would recommend Tan Chai Beng for roles in AI and embedded systems.

[Daniel Kho,CTO, 19/10/2026]

SECTION 8 Verification & Signatures

Student		Supervisor	
Name	Tan Chai Beng	**Name**	Daniel Kho
Signature		**Signature**	
Date	17/10/2026	**Date**	19/10/2026